Experiment 3: Perform Data Transformation Operations in Power BI

Aim

To perform various data transformation operations in Power BI using Power Query Editor in order to clean, modify, and prepare raw data for analysis and visualization.

Theory

Data transformation is the process of converting raw data into a clean and structured format suitable for analysis. In real-world scenarios, collected data is often incomplete, inconsistent, or improperly formatted. Power BI provides a powerful tool called **Power Query Editor** that enables users to perform data cleaning and transformation before loading the data into the data model.

Power Query follows the concept of **ETL (Extract, Transform, Load)**:

1. **Extract** – Import data from various sources such as Excel, CSV, databases, or web sources.
2. **Transform** – Clean and modify the data by applying operations like filtering, sorting, removing duplicates, changing data types, splitting columns, merging tables, etc.
3. **Load** – Load the transformed data into Power BI for reporting and visualization.

Some common data transformation operations in Power BI include:

- **Removing Columns/Rows** – Eliminating unnecessary data.
- **Filtering Rows** – Selecting specific records based on conditions.
- **Sorting Data** – Arranging data in ascending or descending order.
- **Changing Data Types** – Converting columns into appropriate types (text, number, date, etc.).
- **Renaming Columns** – Making column names meaningful and standardized.
- **Handling Missing Values** – Replacing or removing null values.

- **Splitting and Merging Columns** – Dividing a column into multiple columns or combining columns.
- **Grouping Data** – Aggregating data based on categories.
- **Removing Duplicates** – Ensuring data consistency.

Each transformation step is recorded in the **Applied Steps** pane, and Power BI automatically generates M language code in the background. These steps can be modified or deleted without affecting the original data source.

Data transformation improves:

- Data accuracy
- Data consistency
- Report performance
- Quality of insights generated

Thus, proper data transformation is essential before creating dashboards and reports in Power BI.

Procedure

1. **Open Power BI Desktop.**
2. **Import Data:**
   - Click on *Home → Get Data*.
   - Select a data source (e.g., Excel/CSV).
   - Load the dataset.
3. **Open Power Query Editor:**
   - Click on *Transform Data* to open Power Query Editor.
4. **Perform Data Cleaning Operations:**
   - Remove unnecessary columns using *Remove Columns* option.
   - Remove blank rows using *Remove Rows → Remove Blank Rows*.
   - Rename columns by double-clicking on column headers.
   - Change data types from the *Data Type* dropdown.
5. **Apply Filtering and Sorting:**
   - Use filter dropdowns to select specific values.
   - Sort columns in ascending or descending order.
6. **Handle Missing Values:**
   - Replace null values using *Replace Values* option.

    - Or remove rows containing null values.
7. **Split and Merge Columns:**
    - Use *Split Column* (by delimiter) to divide text columns.
    - Use *Merge Columns* to combine multiple columns.
8. **Remove Duplicates:**
    - Select a column → Click *Remove Duplicates*.
9. **Group Data (if required):**
    - Select column → Click *Group By*.
    - Apply aggregation functions such as Sum, Count, or Average.
10. **Close and Apply:**
    - After completing transformations, click *Close & Apply*.
    - The transformed data will be loaded into Power BI.
11. **Create Visualizations:**
    - Use the cleaned dataset to create charts, tables, or dashboards.

Conclusion

In this experiment, various data transformation operations were successfully performed using Power Query Editor in Power BI. The raw dataset was cleaned and modified by removing unnecessary columns, handling missing values, changing data types, filtering records, and removing duplicates.

Data transformation plays a crucial role in ensuring data quality and improving the accuracy of reports and dashboards. Power BI provides a user-friendly and efficient environment for performing ETL operations without altering the original data source. Proper data preparation enhances analytical performance and helps in generating meaningful business insights.

Thus, Power BI's data transformation tools are essential for effective data analysis and visualization.